Transitive string C-groups: the complete half-degree bound

An attributed application of the common string-separator theory

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This application recovers the principal contract of Cameron, Fernandes, Leemans and Mixer, *String C-groups as transitive subgroups of S_n*, arXiv:1410.5863v1, 21 October 2014. Its complete twelve-page edition is fixed. The complete current Topic B common paper and the supplement *String separators, normal modules and transitive rank* are the two declared common inputs. No previous application is required. All source result numbers and qualifications are recorded in the clause map. The finite input is a complete exact census, not a sample of markings.

Theorem 1 (Theorem 1.2, with ordered marking distinctions)**.** *Let G be a finite string C-group of rank d with a faithful transitive action of degree n, with $G \neq A_n, S_n$ in this action. If $d > n/2$, then one of the following occurs:*

1. *$n = 2m$ with m odd, and $G = C_2 \wr S_m$ has rank $m+1$. For $m \geq 5$, its unordered distinguished set is unique up to marked-group isomorphism and is $\{\lambda, \lambda + e_1, s_1, \ldots, s_{m-1}\}$, where λ flips all blocks, e_1 flips the first block, and s_i interchanges blocks $i, i+1$ without internal flips. The two ordered types up to global duality are $[2, 4, 3, \ldots, 3]$ and $[2, 3, \ldots, 3, 4]$.*

2. *G is one of the imprimitive degree-six or degree-eight markings in Table 1 below, with the displayed types and their global reversals.*

3. *$n = 6$, $G \cong S_5$ in its primitive action, and the marking is the rank-four simplex of type $[3, 3, 3]$.*

The first family also exists at $m = 3$ and is included among the degree-six ordered markings. The residual alternative is $d \leq n/2$.

Proof. Choose a maximal nontrivial block system if the action is imprimitive. The common block argument gives $d \leq n/2$ when both block size and number exceed two. For two large blocks of size $k \geq 5$, the common uniform proof and the accepted complete ten-action finite-neighbor lemma give $d \leq k$. For blocks of size two, the common normal-module argument gives $d \leq m + 1$, with equality precisely as in the theorem. It proves the actual marking, cocycle normalization and IP, not merely its order. The complete degree-six/eight/nine census treats all smaller imprimitive actions; odd prime degrees have no imprimitive action, and the proper subgroups of S_4 have independent length at most two. There is no rank-five exception in degree nine.

For primitive actions the common sharp bound and exact residual census give $d < n/2$ except for degree-six A_5, S_5 and degree-ten S_6 with the types printed in the common theorem. Only the degree-six rank-four S_5 marking has $d > n/2$. Its Coxeter relations and order 120 identify it with the simplex. This proves all branches. □

The source's Table 1 prints $[2, 3, 3]$ for the order-36 group. That type is impossible: its universal Coxeter group is $C_2 \times S_4$ of order 48, so no quotient has order 36. The complete exact marking has type $[3, 2, 3]$. This is a correction to this fixed source edition; no claim is made about an unseen later edition. The source's two written types for the wreath family instead reverse only its noncentral connected component. They are compatible orders of the same unordered marked set, but need not be identified as ordered polytopes or as global duals. The present statement includes both.

Degree	Actual group	Order	Ordered types, up to global duality
6	$S_3 \times S_3$	36	$[3,2,3]$
6	$C_2 \times S_4$	48	$[2,3,3]$
6	$C_2 \times S_4 = C_2 \wr S_3$	48	$[2,3,4],\ [2,4,3]$
8	$(2^4 : S_3) : S_3$	576	$[3,4,4,3]$

Table 1: Complete small imprimitive exception types. Literal permutation markings and every ordered conjugacy representative are in the required census.

The common proof establishes Proposition 2.1 for every $n \geq 10$ and Proposition 2.2 in the standing string-C-group setting of source Section 2: when $n \geq 11$, equality $d = n/2 + 1$ forces $n \equiv 2 \pmod 4$ and the unordered pair (G, S) is unique. The complete actual parameter family shows that this bound is sharp. It does not assert that every transitive action of the same abstract group has this degree or rank.

1 The alternating-parabolic consequences

Write R for the parent rank and $r = R - 1$ for a maximal parabolic rank. The source uses $S = \{\rho_0, \ldots, \rho_d\}$ in Section 4, so there its printed d is r, not R.

Corollary 1 (Source Corollaries 4.1–4.3)**.** *Let $G = A_n$ with a string C-marking of rank R, and let G_i be a maximal parabolic in its natural degree-n action. If G_i is transitive imprimitive, then $r \leq n/2$, hence $R \leq \lfloor n/2 \rfloor + 1$; in particular the source's literal $r \leq n/2 + 1$ holds. If G_i is primitive, then $r < n/2$, hence $R \leq \lceil n/2 \rceil$ and the source's literal $r \leq n/2$ holds. Consequently if $R \geq n/2 + 2$, every maximal parabolic is intransitive.*

Proof. A parabolic is proper by IP, so it is not A_n or S_n. The imprimitive equality family has the odd permutation λ when m is odd, and therefore cannot lie in A_n. All degree-six rank-four markings likewise contain an odd permutation; the census checks this on every actual tuple. The only remaining exception is the degree-eight rank-five marking in A_8. Its parent would have rank six in A_8. The complete rank-six connected-string census in A_8 is empty. Since A_8 is directly indecomposable and centerless, its generating C-diagram must be connected. This rules out that case and proves $r \leq n/2$.

For a primitive parabolic the degree-six S_5 and degree-ten S_6 actions are not contained in the corresponding alternating group, as the explicit permutation models verify. The only remaining half-degree case is A_5 in degree six, of rank three. Its parent would be a rank-four C-group marking of A_6, which the same complete census excludes. Hence $r < n/2$. The final assertion follows because $R \geq n/2 + 2$ would give $r > n/2$. □

The rank convention is essential: in degree five the icosahedral A_5 marking has parent rank three and a transitive primitive D_{10} parabolic. It satisfies the source's indexed inequality $r = 2 \leq 5/2$, but not the incorrectly relabelled assertion $R \leq 5/2$.

2 Scope and attribution

The source's introductory alternating-group Conjecture 1.1 is attributed to its earlier authors. The already accessible paper *Highest rank of a polytope for A_n* (catalogue L044) states that it resolves that conjecture. It is not a new solution or an open-question claim here. The introductory symmetric-group and intransitive-product examples are supplied by the declared common extremal theorem and the elementary direct-product construction. Maróti's theorem is retained

in its original precise form. The finer Table 3 maximum-rank reports remain attributed ancillary reports, except the explicitly recovered entries in the clause map; none is a premise of the principal proof.